DAWSON CITY WWTP SYSTEM UPSET

PREPARED BY

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PREPARED FOR

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INTRODUCTION

AnthonyClarke

Yukon Territory

Dear Anthony,

ACC Wastewater Solutions is pleased to present our proposal for the **Dawson City WWTP System Upset .**

I would like to thank you for giving ACC Wastewater Solutions the opportunity to help you develop a long-term mediation plan for the Yukon Territory's Wastewater System. I appreciate your time helping us understand your system and the concerns you have.

At ACC Wastewater Solutions we are committed to providing our customers with cost-effective tools to identify and manage the complex problems that may be preventing our customers from achieving their treatment target. We're committed to:

1. Exceptional customer service

- Letting science "lead the way"
- Total client transparency (even if it means the lack of sale, or admitting when we don't have the answer yet)
- Providing targeted solutions using industry-leading treatment products and technologies

I hope that you find this proposal helpful and look forward to speaking with you about it in the near future.

Sincerely,

Raymond Menard

ACC Wastewater Solutions

At ACC Wastewater Solutions we've been working with pond water, lake water, stormwater, raw water and wastewater for nearly 25 years. Since 1998, we've been managing our very own wastewater treatment plant in the form of a recirculating aquaculture facility, Smoky Trout Farm. We keep our rainbow trout alive and healthy in a plant that recycles more than 99% of its water. We understand what it's like when things go right, but also know what it's like when things go wrong.

"Follow the science" is our mantra. This allowed us to understand and overcome issues in our facility, and it's that same one that allows us help our clients achieve their treatment targets in their systems too.

ACC Wastewater Solutions is a member of the Alberta Water and Wastewater Operators Association, the Saskatchewan Water and Wastewater Association, the Manitoba Water and Wastewater Association, the North American Lake Management Society, the Alberta Lake Management Society, Western Canada Water and other industry organizations.

OUR TEAM

DHAVALKUMAR VAGHASIYA ENVIRONMENTAL SCIENTIST

After completing his Bachelors in Chemical Engineering, Dhaval was able to narrow down his interest encircling environmental studies and its disciplines, and went ahead to complete his Master's in Environmental Management, where he was successful in publishing a research paper on the Treatment of CETP Effluent by Green Oxidant Ferrate - an alternative to the primary treatment in his term.

Further interest brought him to Canada for a Master's in Energy and Environment Engineering, which helped him polish his skills in designing and assessing the environmental impact of engineering plans. His strengths lie in aeration designing, wastewater treatment and recycling, soil remediation and recently acquired skill of bathymetric survey.

MAX MANARD AERATION DESIGN | FISHERIES MANAGEMENT

Max founded Smoky Trout Farm with his father, Dan, more than two decades ago. His water management knowledge is based in the extensive management of our own Recirculating Aquaculture Facility. There are few people in Canada with as much experience as Max at improving and maintaining private recreational water bodies.

In partnership with his brother Ray, Max launched the Algae Control Canada brand in 2016. Their goal; to offer water management solutions to customers beyond fisheries management. In addition to the management of fish-bearing waters, Max has collaborated on the design of aeration systems and treatment programs for stormwater ponds, raw water reservoirs, and wastewater lagoons.

RAY MANARD MANAGING PARTNER | PROJECT MANAGER

After two decades in the financial service industry, Ray joined the family business in 2014. His experience as a commercial banker and high-level relationship manager gave him a unique perspective on the water management business. As a business banker, Ray recognized there was no way he could be an expert in everything his clients needed, so he gained an in-depth knowledge of business issues and developed a network of professional partners, including lawyers, investment managers, accountants and insurance specialists so he could leverage and ensure his customers were getting the best advice possible for their situation. Ray applies exactly the same principle to the water management business.

Through membership in industry associations and attendance at their conferences and seminars, Ray has developed a familiarity with the many water quality issues we face today and broad knowledge of the tools that are available to manage them. More importantly, the relationships he has cultivated with top limnologists, aquatic scientists, biologists, chemists and engineers give ACC Wastewater Solutions access to an unparalleled knowledge base.

Eric Fehrman SALES MANAGER

Eric comes from an 18 year background of management, operations, marketing and relationship development. He has always placed a large focus on customer service, quality control and client satisfaction to build strong relationships with clientele. Most recently, he spent 3yrs of his career as a National Account Manager, overseeing municipal, provincial and federal levels of government. He was involved in complex negotiations and project management on large-scale capital projects domestically, as well as high-level security projects throughout the world. A significant portion of the projects he was involved in were within water/wastewater and working on some of the largest wastewater facilities within the country, which is what led to him becoming part of the ACC team.

SOME OF OUR PARTNERS

One of the biggest barriers to success is trying to be all things to all people. At ACC Wastewater Solutions we're good at what we do, but when we run into something we haven't seen before, we're not afraid to say "We don't know." But even if we don't know the solution to a problem, we almost certainly know someone who does.

KEVIN RIPP DIRECTOR OF SCIENCE AND INNOVATION | AQUAFIX

DÖRTE KÖSTER, PH.D. SENIOR AQUATIC SCIENTIST | ASSOCIATED ENVIRONMENTAL CONSULTANTS

PATRICK GOODWIN RESEARCH SCIENTIST | VERTEX AQUATIC SOLUTIONS

YOUR PROCESS

INFORMATION AT GLANCE

- Flow rate of 3000 m3 (792,516 GPD)

Identified Issues

- Had major convergence event
- Bacteria has been inactive since the event and have never come back
- Bacteria levels are flushed out in winter
- Water goes down to 5 degrees at the lowest with sluggish bacteria
- All aerobic in the plant, but the event turned everything anaerobic (septic)

DISCUSSION LOW TEMPERATURE BIOLOGICAL ACTIVITY

Biological activity in wastewater treatment plants can be affected by various factors, including temperature. In the winter, several key factors come into play that can impact the performance of biological processes in wastewater treatment:

1. **Reduced Microbial Activity**: Cold temperatures can slow down the growth and activity of the bacteria and other microorganisms that are responsible for biodegrading organic pollutants in wastewater. This can result in longer retention times and reduced treatment performance.
2. **Temperature**: Microbial activity is highly temperature-dependent. In colder winter months, the water temperature in wastewater treatment plants can drop significantly, which can slow down the metabolism of microorganisms responsible for breaking down organic matter in the wastewater. This can lead to reduced treatment efficiency.
3. **Sludge Accumulation**: Cold temperatures can lead to increased sludge accumulation in treatment tanks, which may reduce the effective volume available for wastewater treatment. This can also result in lower treatment efficiency and increased maintenance requirements.
4. **Increased Energy Costs**: To maintain a suitable temperature for biological processes during the winter, treatment plants may need to invest in heating systems, which can increase energy consumption and operational costs.

To mitigate these issues, wastewater treatment plants in colder climates often employ various strategies, such as:

1. **Use of Cold-Tolerant Microbes**: In some cases, treatment plants may use cold-tolerant microbial cultures that can remain active at lower temperatures.
2. **Adjusting Process Control**: Operators may need to make adjustments to process control parameters, such as hydraulic retention time and mixed liquor suspended solids (MLSS) concentrations, to optimize treatment performance during winter.
3. **Insulation**: Insulating treatment tanks and other infrastructure can help maintain a more stable temperature, reducing the impact of winter cold on biological activity.
4. **Heat Exchangers**: Some plants use heat exchangers to capture and reuse heat generated during treatment processes to maintain a suitable temperature for microbial activity.
5. **Extended Aeration**: Extending the aeration time in activated sludge processes can give microorganisms more time to break down organic matter, compensating for reduced metabolic rates in colder conditions.

Overall, winter conditions can pose challenges to biological wastewater treatment processes. Still, with careful planning, operational adjustments, and appropriate technology, treatment plants can continue to treat wastewater during the colder months effectively.

DISCUSSION RECOVERING FROM SYSTEM UPSET

Operators want to avoid a toxicity-induced wastewater system upset. It's easy to understand why. A system upset comes with real problems like:

- a rise in pH
- an increase in TSS
- an upward spike in ammonia levels
- a drop in dissolved oxygen levels, and
- more sludge production

For operators when you see the signs of a system upset your best response is to rebuild your healthy bacteria base. You can increase its vitality and recreate the conditions of a healthy and functioning process.

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WHAT IS THE CAUSE?

More toxins in your influent are the top reason for a system upset. The chemicals designed to kill harmful bacteria in health-care, food production and home disinfection can make their way into the wastewater system. Once these toxic chemicals are in your wastewater influent, they continue to destroy both harmful and beneficial bacteria.

Other events that trigger system upsets:

- a change in temperature,
- low food (offseason in a vacation town) or
- seasonal slug loads of fats, oils and grease (FOGs).

HOW TO SOLVE A WASTEWATER SYSTEM UPSET?

Once toxins have entered your wastewater system you will be able to see the visible signs of beneficial bacteria diminishing. Symptoms like less floc and an increase in TSS. The best response is to rebuild bacteria.

PROPOSAL

FOR INCREASING BIOLOGICAL ACTIVITY

ACC Wastewater Solutions recommends a treatment cycle with **VitaStim Polar** to promote biological activity in wastewater treatment plants during winter months. The probiotic blend in **VitaStim Polar** promotes healthy biological activity in both the indigenous and added bacteria, and reduces compacted sludge, clarifies the water, and reduces odors. **VitaStim Polar** saves systems during the dead of winter because of its ability to degrade municipal wastes, phenols, surfactants and other industrial wastes in temperatures as low as 41° F.

FOR RECOVERING FROM SYSTEM UPSET

SmartBOD is an advanced carbon source that provides the building blocks for biological life. This product offers a cost-effective solution for chronic issues related to low food and flows that are too low for the plant design.

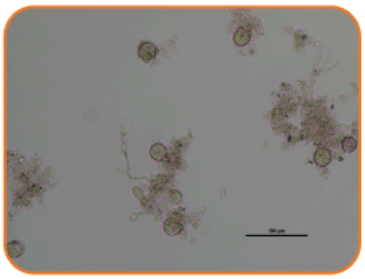
Wastewater plants fed with **SmartBOD** have better-settling properties and produce clearer effluent than plants fed with other carbon supplements. Also, plants with a history of low F:M filaments, like Type 1851 and Type 0041/0675, become less prone to filament outbreaks.

Food supplement products like dog food and chicken feed have traditionally been used to bolster BOD. These foods contain higher levels of fats and phosphorus than are beneficial to wastewater bacteria. They also contain high levels of inert solids which will accumulate in a wastewater treatment system and trap organic matter, leading to low DO conditions.

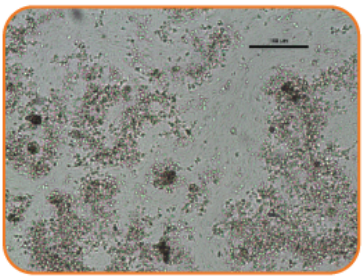
Crude glycerin and molasses provide a food supplement in the form of sugar. While they can grow a population of bacteria quickly, they are not well equipped to digest anything other than sugar and provide little benefits digesting common municipal or industrial BOD. In fact, sugar based additives typically result in a cloudy effluent and polysaccharide slime.

The human body cannot run on sugar alone and neither can a wastewater treatment plant. Traditional food supplements may offer a source of fuel, but they lack the nutrient balance and floc building components wastewater bacteria need. **SmartBOD** is a complete source of fuel to improve your biology at a significantly lower dosage than sources like dog and chicken feed or a source of sugar.

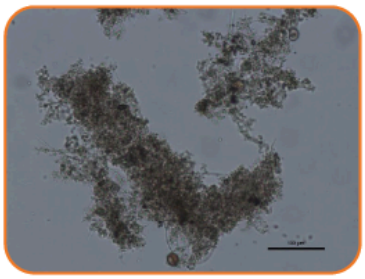
Floc formation before and after SmartBOD treatment



Day 1:
Poor nutrient balance resulted in only partial floc formation.



Day 7:
After adding SmartBOD for seven days, bacterial health is improving.



Day 14:
Flocs are fully formed and settling well.

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PRODUCT INFORMATION

VITASTIM POLAR

VitaStim Polar is a potent blend of 23 essential vitamins, biostimulants, biocatalysts, and select biological species custom-blended for wastewater treatment plants to promote biological activity during winter months

BENEFITS

- Creates a healthier biomass under cold temperatures
- Promotes nitrification
- Provides great recovery from upsets
- Increases cell metabolism during the depths of winter
- Improves settling

PRODUCT INFORMATION

SMARTBOD

SmartBOD is an advanced carbon source that provides the building blocks for biological life. This product offers a cost-effective solution for chronic issues related to low food and flows that are too low for the plant design. SmartBOD contains bioavailable amino acids, complex proteins, carbohydrates, and micronutrients that bacteria in biological wastewater processes require in order to build floc and effectively remove nutrients.

BENEFITS:

- Balanced blend of nutrients to promote flocculation and improve settling
- Boosts health of bacteria population
- Proven to outperform glycerin and dog food
- Great for systems with low or intermittent loading
- Lowers effluent, TSS, ammonia, and phosphorous while reducing bulking
- Better BOD and nutrient removal

PRODUCT APPLICATION RATES - 1-Month

VITASTIM POLAR

- 8lbs applied per day for the first 10 days
- 4lbs applied per day for continued maintenance dose

PRODUCT APPLICATION RATES - 1-Month

SMARTBOD - RECOMMENDED DOSE BUT CAN BE USED AS NEEDED

- 7 day schedule: 12lbs applied 7 days/week
- 5 day schedule: 17lbs applied 5 days/week
- You can use it as needed in case your DO level increase, as well as play around with the dosing

Description	Price	Qty	Subtotal
Vitastim Polar 30lb Container, 1lb Water Soluble Pks	$820	5	$4,100
SmartBOD Bag - 22.68kg (50LB) Bulk	$520	5	$2,600
Shipping TBD			$0
TOTAL			**$6,700**

Proposal Sign-off

Algae Control Canada looks to establish a collaborative and sustainable relationship with our customers that delivers timely, creative, effective, and budget-conscious solutions. We look forward to building a relationship with the Yukon Territory.

ACC Wastewater Solutions proposes to provide all of the recommended products described above for a fee of $6,700.00 plus shipping and taxes. This price includes all products and services presented above. We look forward to working alongside you this year!

Payment options are available.

The budget is in Canadian Dollars and excludes sales tax and shipping. Upon your approval of the recommendations and pricing in this proposal (including any negotiated amendments), a sales order will be forwarded to you for final approval.

SIGNATURE
Raymond Menard

Raymond Menard

ACC Wastewater Solutions

SIGNATURE
Anthony Clarke

Anthony Clarke

Yukon Territory